

# Reproducible Research: A Valid Concept?

Nathaniel E. Helwig

*Assistant Professor of Psychology and Statistics*  
University of Minnesota (Twin Cities)



MRI Users' Group Talk  
April 12, 2016

# Defining Reproducible Research

**Reproducible research** is the idea that data analyses, and more generally, scientific claims, are published with their data and software code so that others may verify the findings and build upon them.

<https://www.coursera.org/learn/reproducible-research>

The goal of **reproducible research** is to tie specific instructions to data analysis and experimental data so that scholarship can be recreated, better understood and verified.

<https://cran.r-project.org/web/views/ReproducibleResearch.html>

The term **reproducible research** refers to the idea that the ultimate product of academic research is the paper along with the full computational environment used to produce the results in the paper such as the code, data, etc. that can be used to reproduce the results and create new work based on the research.

<https://en.wikipedia.org/wiki/Reproducibility>

# Outline of Talk

## 1) Reproducibility vs Validity

- Motivating example
- Quest for validity

## 3) Reproducible Components

- Component models
- Example with fMRI data

## 2) Reproducible Relationships

- Nonparametric regression
- Example with EEG data

# Reproducibility vs Validity

# Results From Four Hypothetical Studies

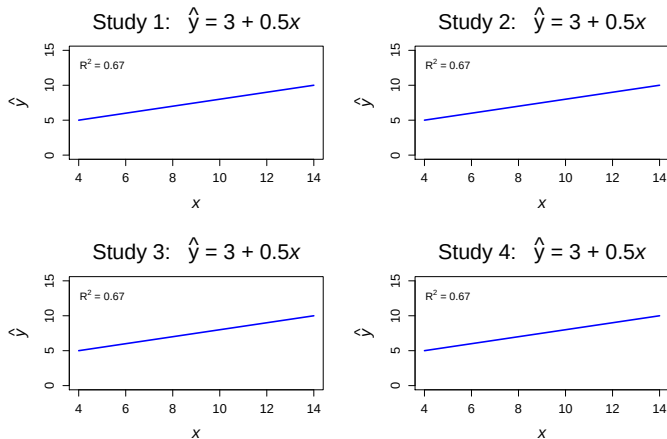


Figure 1: Estimated linear relationship from four hypothetical studies.

# Implications of Four Hypothetical Studies

What do the results on the previous slide imply?

Can we conclude that there is a reliable linear relationship between  $X$  and  $Y$ ?

Is the reproducibility of the finding indicative of a valid discovery?

What have we learned about the data from these results?

# Let's Look at the Data

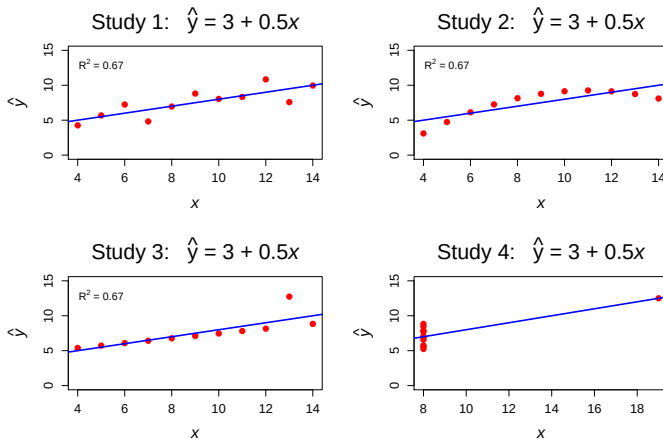


Figure 2: Estimated linear relationship with corresponding data.

# Anscombe's (1973) Quartet

```
> anscombe
```

|    | x1 | x2 | x3 | x4 | y1    | y2   | y3    | y4    |
|----|----|----|----|----|-------|------|-------|-------|
| 1  | 10 | 10 | 10 | 8  | 8.04  | 9.14 | 7.46  | 6.58  |
| 2  | 8  | 8  | 8  | 8  | 6.95  | 8.14 | 6.77  | 5.76  |
| 3  | 13 | 13 | 13 | 8  | 7.58  | 8.74 | 12.74 | 7.71  |
| 4  | 9  | 9  | 9  | 8  | 8.81  | 8.77 | 7.11  | 8.84  |
| 5  | 11 | 11 | 11 | 8  | 8.33  | 9.26 | 7.81  | 8.47  |
| 6  | 14 | 14 | 14 | 8  | 9.96  | 8.10 | 8.84  | 7.04  |
| 7  | 6  | 6  | 6  | 8  | 7.24  | 6.13 | 6.08  | 5.25  |
| 8  | 4  | 4  | 4  | 19 | 4.26  | 3.10 | 5.39  | 12.50 |
| 9  | 12 | 12 | 12 | 8  | 10.84 | 9.13 | 8.15  | 5.56  |
| 10 | 7  | 7  | 7  | 8  | 4.82  | 7.26 | 6.42  | 7.91  |
| 11 | 5  | 5  | 5  | 8  | 5.68  | 4.74 | 5.73  | 6.89  |

# Why Reproducibility is Not Enough

Having a reproducible finding is good, but we should strive for more!

Research findings can be reproducible for different reasons

- ➊ Due to genuine features of data (what we want)
- ➋ Due to biases we impose on data (what we want to avoid)

How can we encourage the former and avoid the latter?

- There is no simple solution to guarantee genuine findings
- But there are statistical solutions to encourage them

# Thinking Statistically about Data

Suppose we have some data  $\mathbf{x}$  that follows some distribution  $F$

- Example:  $\mathbf{x} = (x_1, \dots, x_n)$  are iid from  $N(\mu, \sigma^2)$

Let  $\theta = t(F)$  denote some **parameter** of interest about the distribution  $F$

- The notation  $t(\cdot)$  denotes that  $\theta$  is a function of  $F$
- Example:  $\mu = \int xf(x)dx$  is the expected value

Let  $\hat{\theta} = s(\mathbf{x})$  denote some **statistic** used to estimate the parameter  $\theta$

- The notation  $s(\cdot)$  denotes that  $\hat{\theta}$  is a function of  $\mathbf{x}$
- Example:  $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$  is the sample mean

Want to make inferences about the parameter  $\theta$  using the statistic  $\hat{\theta}$ .

# Bias, Variance, and Mean-Squared Error

How can we judge the quality of  $\hat{\theta}$  and corresponding inferences about  $\theta$ ?

The **bias** of an estimate  $\hat{\theta} = s(\mathbf{x})$  of  $\theta = t(F)$  is defined as

$$\text{Bias}_F = E_F[s(\mathbf{x})] - t(F)$$

where the expectation is taken with respect to  $F$ .

The **mean-squared error (MSE)** of an estimate  $\hat{\theta} = s(\mathbf{x})$  of  $\theta = t(F)$  is

$$\begin{aligned}\text{MSE}_F &= E_F\{[s(\mathbf{x}) - t(F)]^2\} \\ &= V_F(\hat{\theta}) + [\text{Bias}_F(\hat{\theta})]^2\end{aligned}$$

where the expectation is taken with respect to  $F$ .

# Balance between Accuracy and Precision

MSE quantifies both the accuracy (bias) and precision (variance).

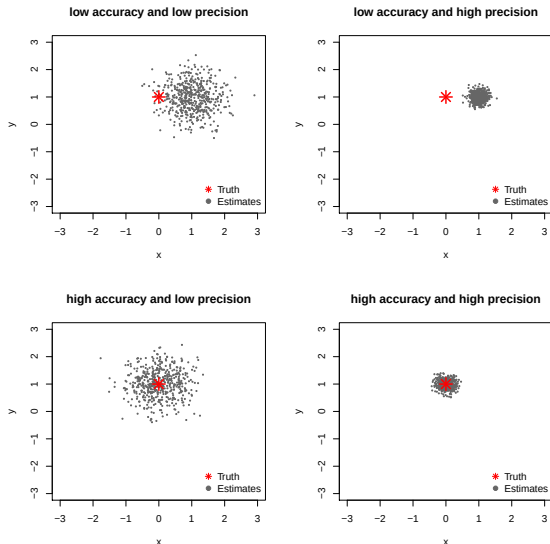
Ideal estimators are accurate (small bias) and precise (small variance).

Having some bias can be an ok (or even good) thing, despite the negative connotations of the word “biased”.

Would you rather have an estimator that is biased by 1 unit with a standard error of 1 unit? Or one that is unbiased but has a standard error of 1.5 units?

- In this case, the biased estimator is better with respect to MSE

# Accuracy and Precision Visualization



**Figure 3:** Depiction of estimators with different statistical characteristics. Worst scenario is top left, whereas ideal scenario is bottom right.

# Reproducible Relationships

# Parametric versus Nonparametric Regression

The general linear model is a form of **parametric regression**, where the relationship between  $Y$  and  $X_1, \dots, X_p$  has some predetermined form.

- Parameterizes relationship between  $Y$  and  $X_1, \dots, X_p$
- Then estimates the specified parameters
- Great if you know the form of the relationship (e.g., linear)

In contrast, **nonparametric regression** tries to estimate the form of the relationship between  $Y$  and  $X_1, \dots, X_p$ .

- No predetermined form for relationship between  $Y$  and  $X_1, \dots, X_p$
- Great for discovering relationships and building prediction models

# Finding Functional Relationships in Noisy Data

A typical nonparametric regression model has the form

$$y_i = \eta(\mathbf{x}_i) + \epsilon_i$$

where  $\eta(\cdot)$  is an unknown function relating the predictors  $\mathbf{x}_i = (x_{i1}, \dots, x_{ip})'$  and the response  $y_i$ , and  $\epsilon_i \stackrel{\text{iid}}{\sim} F(0, \sigma^2)$  are model error terms.

Goal is to estimate  $\eta$  from the observed data.

- Many approaches exist (see Gu, 2013; Hastie & Tibshirani, 1990; Hastie et al., 2009; Ruppert et al., 2003; Wahba, 1990; Wood, 2006)
- Need to find balance between model fit and model parsimony

# Example of Nonparametric Regression

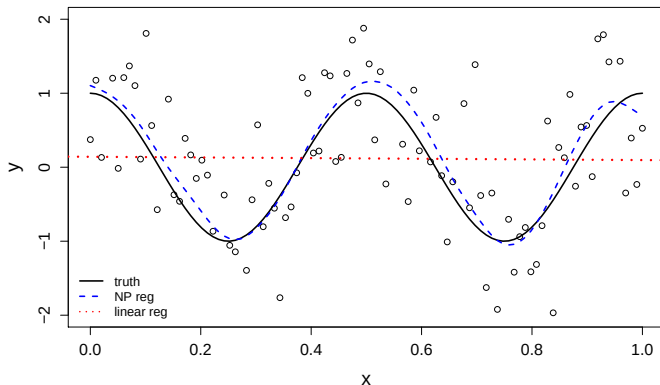


Figure 4: True function is  $\eta(x) = \cos(4\pi x)$ . NP reg is cubic smoothing spline.

# Simple Simulation Study Design

Linear regression (`lm.R`) vs smoothing spline (`smooth.spline.R`)

- Four examined functions (see below)
- Four examined sample sizes ( $n \in \{100, 200, 500, 1000\}$ )

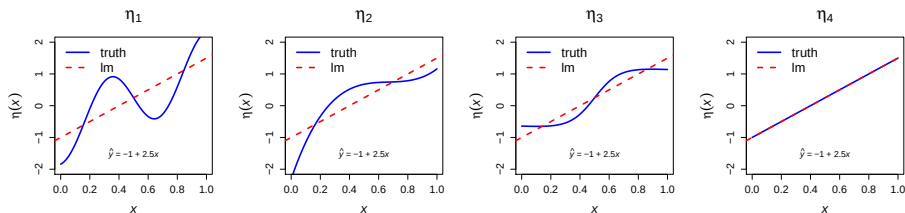
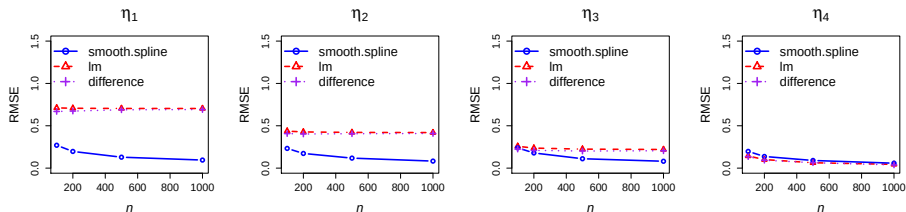


Figure 5: True functions along with best linear approximation.

# Simple Simulation Study Results

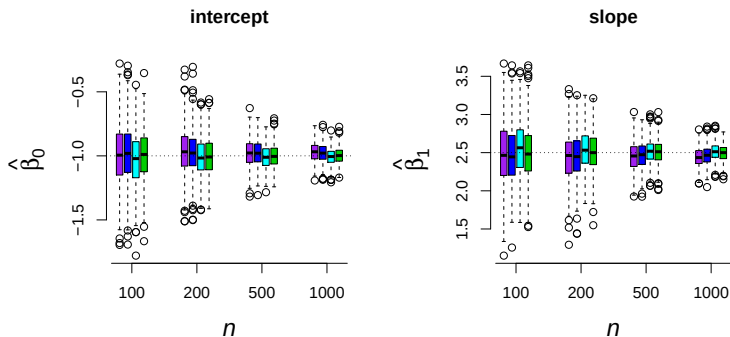
Note that the smoothing spline does well in all conditions, but the linear regression model only does well for  $\eta_4$  (the linear function).



**Figure 6:** Root mean-squared error (RMSE) between estimates and truth (for smooth.spline and lm) and between smooth.spline and lm estimates (difference).

## Simple Simulation Study Results (continued)

Note the best fitting linear regression coefficients ( $\beta_0 = -1$  and  $\beta_1 = 2.5$ ) are reliably recovered as  $n$  increases (but these coefficients are only valid for  $\eta_4$ ).



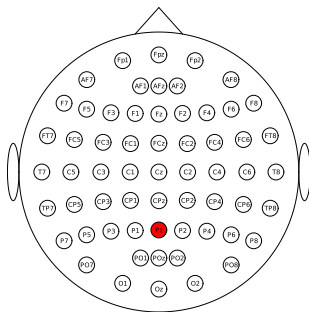
**Figure 7:** Boxplots of estimated intercept and slope from linear regression models. Purple box is  $\eta_1$ , blue box is  $\eta_2$ , cyan box is  $\eta_3$ , and green box is  $\eta_4$ .

# Visual Stimulus ERP Study

Data from UCI Machine Learning Repository; originally collected by **Henri Begleiter** at the Neurodynamics Laboratory at SUNY Brooklyn.

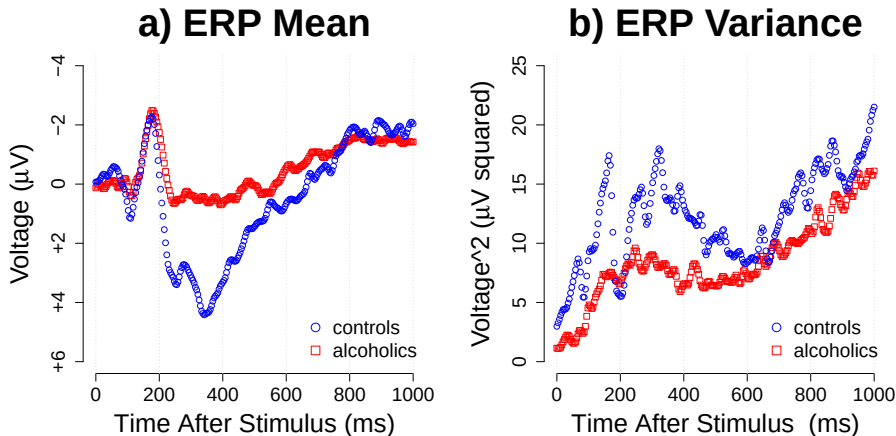
## Experiment Details:

- Data from 120 subjects
  - 76 alcoholics
  - 44 controls
- 64-channel EEG cap
  - Ground: Nz
  - Reference: Cz
- Snodgrass & Vanderwart
  - Set of 260 pictures
- Many trials sampled at 256 Hz for one second after stimulus



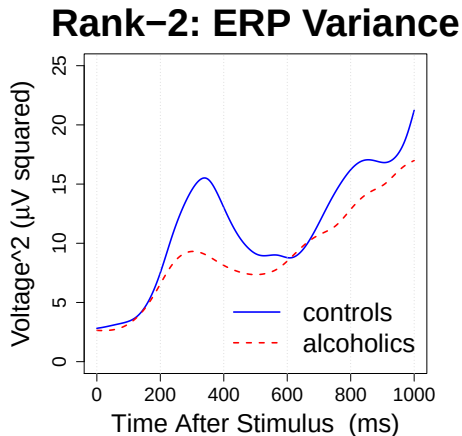
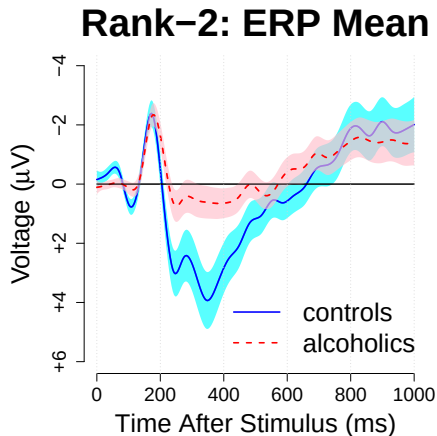
**Figure 8:** The 64-channel EEG cap.

# Observed Mean and Variance of ERP Signal



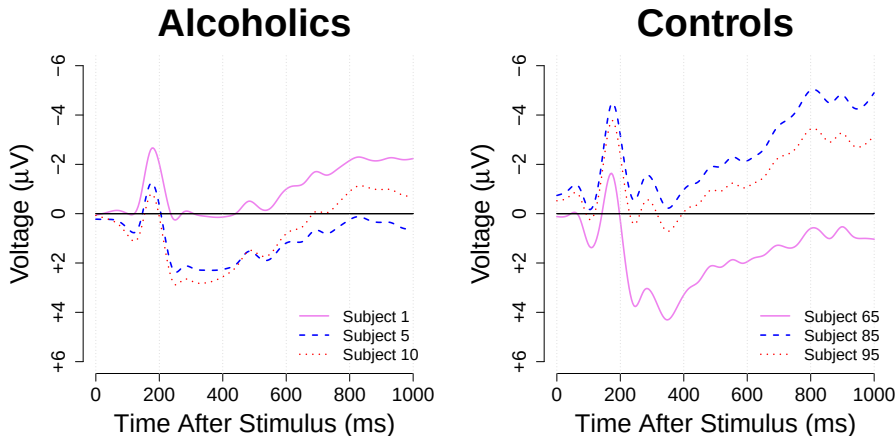
**Figure 9:** Observed mean and variance for electrode Pz. From Helwig (in press).

# Estimated Mean and Variance of ERP Signal



**Figure 10:** Estimated mean and variance for electrode Pz. From Helwig (in press).

# Individual Differences in the ERP Signal



**Figure 11:** Estimated mean and variance for electrode Pz. From Helwig (in press).

# Reproducible Components

# Bilinear Model Form (PCA, FA, ICA)

Let  $\mathbf{X} = \{x_{ij}\}_{I \times J}$  where  $x_{ij}$  is  $i$ -th subject's observed value on the  $j$ -th variable.

A **bilinear model** assumes that

$$\mathbf{X} = \mathbf{A}\mathbf{B}' + \mathbf{E} \quad \longleftrightarrow \quad x_{ij} = \sum_{r=1}^R a_{ir}b_{jr} + e_{ij} \quad (1)$$

where

- $\mathbf{A} = \{a_{ir}\}_{I \times R}$  with  $a_{ir}$  denoting the weight (**score**) of the  $i$ -th subject on the  $r$ -th factor/component
- $\mathbf{B} = \{b_{jr}\}_{J \times R}$  with  $b_{jr}$  denoting the weight (**loading**) of the  $j$ -th variable on the  $r$ -th factor/component
- $\mathbf{E} = \{e_{ij}\}_{I \times J}$  with  $e_{ij}$  denoting the error term corresponding to  $x_{ij}$

# Rotational Indeterminacy Problem

Suppose that  $\mathbf{R}$  is an  $R \times R$  orthogonal rotation matrix.

- $\mathbf{R}'\mathbf{R} = \mathbf{R}\mathbf{R}' = \mathbf{I}$  (identity matrix)

Rotational indeterminacy problem of the bilinear model:

$$\mathbf{AB}' = \tilde{\mathbf{A}}\tilde{\mathbf{B}}'$$

where  $\tilde{\mathbf{A}} = \mathbf{AR}$  and  $\tilde{\mathbf{B}} = \mathbf{BR}$ .

Need to make some assumptions to solve the rotational indeterminacy.

- PCA assumes components are orthogonal and explain maximal variance
- ICA assumes components are statistically independent

# Two-Way versus Three-Way Arrays

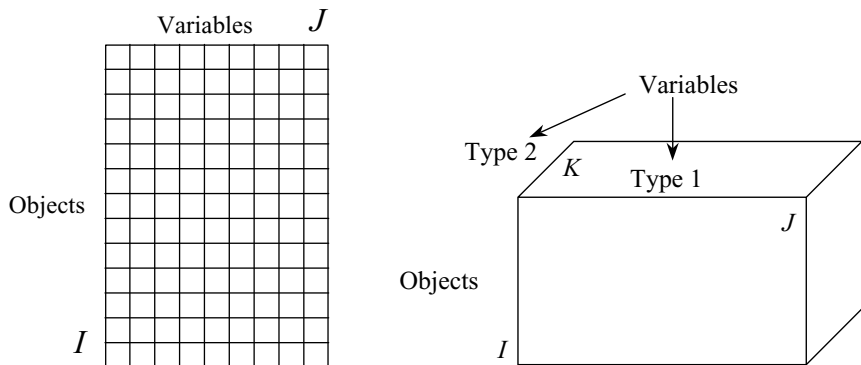


Figure 12: Visualization of 2-way and 3-way arrays from Smilde, Bro, and Geladi (2004)

.

# “The Covariation Chart” from Cattell (1952)

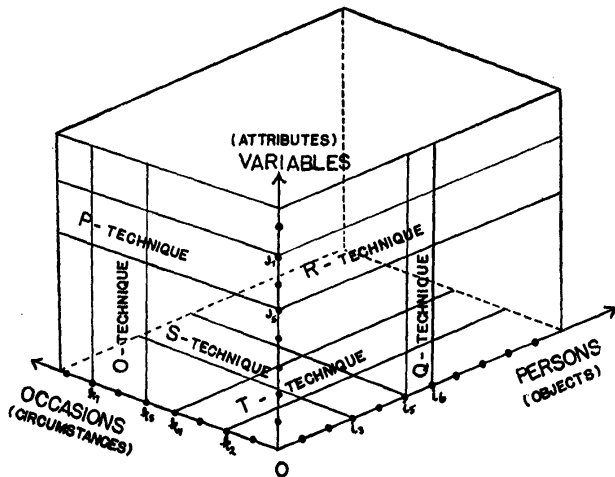


Figure 13: The first illustration of a three-way array.

# Tucker's (1966) Three-Way Factor Analysis Model

$$x_{ijk} = \sum_{r=1}^R \sum_{s=1}^S \sum_{t=1}^T g_{rst} a_{ir} b_{js} c_{kt} + e_{ijk}$$

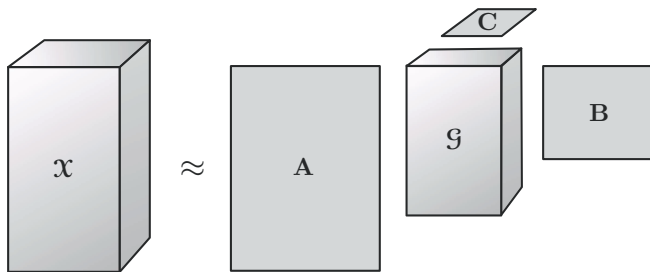
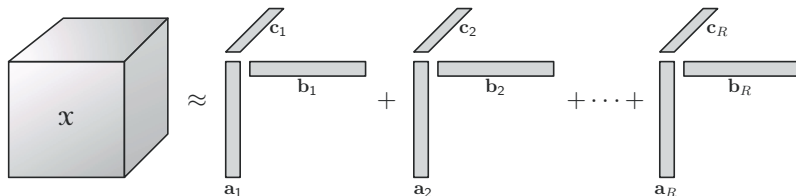


Figure 14: Visualization of Tucker3 structure from Kolda and Bader (2009).

# A Popular Model for Three-Way Data

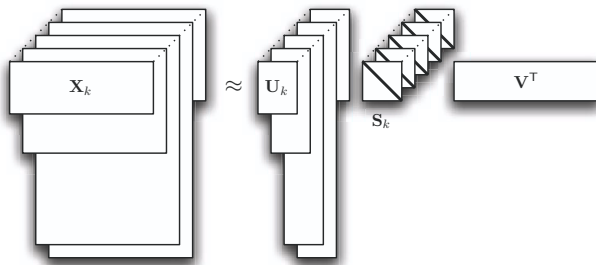
$$x_{ijk} = \sum_{r=1}^R a_{ir}b_{jr}c_{kr} + e_{ijk}$$

| Name                                        | Proposed by                  |
|---------------------------------------------|------------------------------|
| Polyadic form of a tensor                   | Hitchcock, 1927 [105]        |
| PARAFAC (parallel factors)                  | Harshman, 1970 [90]          |
| CANDECOMP or CAND (canonical decomposition) | Carroll and Chang, 1970 [38] |
| Topographic components model                | Möcks, 1988 [166]            |
| CP (CANDECOMP/PARAFAC)                      | Kiers, 2000 [122]            |



**Figure 15:** Visualization of trilinear structure from Kolda and Bader (2009).

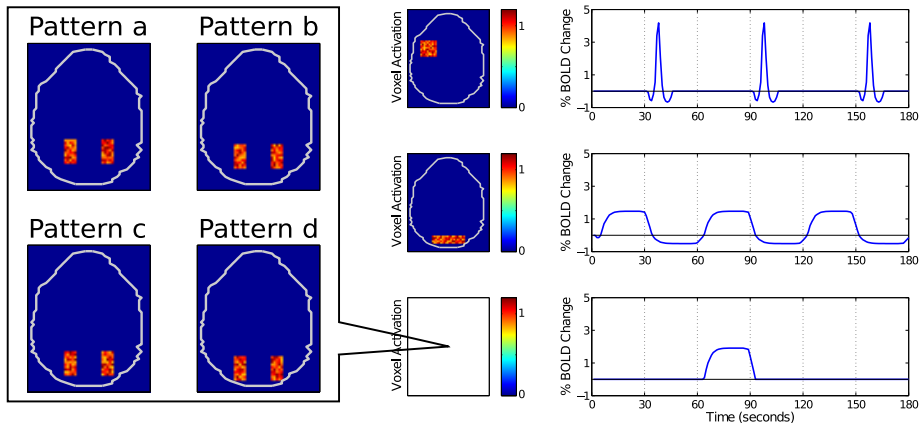
# Simultaneous Component Analysis Models



**Figure 16:** Visualization of SCA structure from Kolda and Bader (2009).

The Parafac (Harshman, 1970) and Pararac2 (Harshman, 1972) models are special cases of the SCA framework (Timmerman & Kiers, 2003).

# Some Simulated fMRI Data (Helwig & Hong, 2013)



Spatial pattern of activity differs across four patterns:

- (a) No overlapping activity
- (b) 10% overlapping activity
- (c) 30% overlapping activity
- (d) 50% overlapping activity

# Parafac versus TPICA (SNR=0.10)

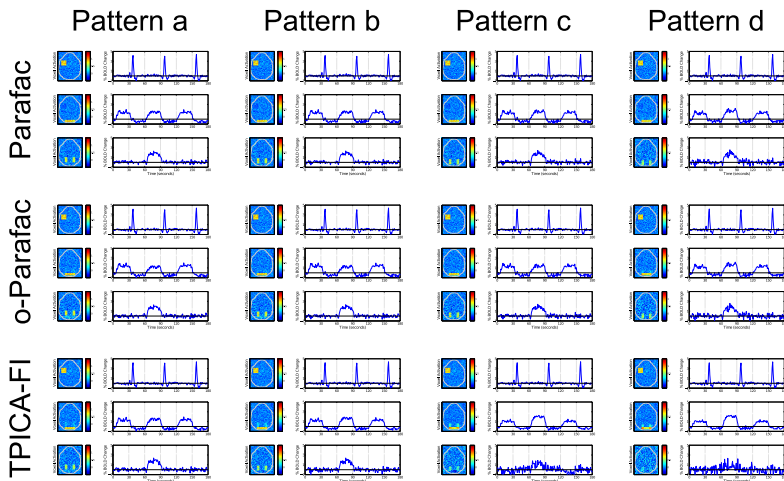


Figure 17: Estimated voxel maps and time courses. From Helwig and Hong (2013).

# Take Home Message

If reproducible research is due to genuine features of data, the results should be obtainable using multiple different methodologies.

If reproducible research is due to biases we impose on data, the results should differ depending on the used methodology.

Reproducing a result via the same method does NOT imply the result is valid.

To ensure reproducible research is valid, we should rely on converging evidence from multiple different methodologies.

# References

- Anscombe, F. J. (1973). Graphs in statistical analysis. *American Statistician* 21, 17–21.
- Cattell, R. B. (1952). The three basis factor-analytic research designs — their interrelations and derivatives. *Psychological Bulletin* 49, 499–520.
- Gu, C. (2013). *Smoothing Spline ANOVA Models* (Second ed.). New York: Springer-Verlag.
- Harshman, R. A. (1970). Foundations of the PARAFAC procedure: Models and conditions for an “explanatory” multimodal factor analysis. *UCLA Working Papers in Phonetics* 16, 1–84.
- Harshman, R. A. (1972). Parafac2: Mathematical and technical notes. *UCLA Working Papers in Phonetics* 22, 30–44.
- Hastie, T. and R. Tibshirani (1990). *Generalized Additive Models*. New York: Chapman and Hall/CRC.
- Hastie, T., R. Tibshirani, and J. Friedman (2009). *The Elements of Statistical Learning*. New York: Springer-Verlag.
- Helwig, N. E. (in press.). Efficient estimation of variance components in nonparametric mixed-effects models with large samples. *Statistics and Computing*.
- Helwig, N. E. and S. Hong (2013). A critique of Tensor Probabilistic Independent Component Analysis: Implications and recommendations for multi-subject fMRI data analysis. *Journal of Neuroscience Methods* 213, 263–273.
- Kolda, T. G. and B. W. Bader (2009). Tensor decompositions and applications. *SIAM Review* 51, 455–500.
- Ruppert, D., M. P. Wand, and R. J. Carroll (2003). *Semiparametric Regression*. Cambridge: Cambridge University Press.
- Smilde, A., R. Bro, and P. Geladi (2004). *Multi-way analysis with applications in the chemical sciences*. West Sussex: John Wiley & Sons.
- Timmerman, M. E. and H. A. L. Kiers (2003). Four simultaneous component analysis models for the analysis of multivariate time series from more than one subject to mode intraindividual and interindividual differences. *Psychometrika* 68, 105–121.
- Tucker, L. R. (1966). Some mathematical notes on three-mode factor analysis. *Psychometrika* 31, 279–311.
- Wahba, G. (1990). *Spline models for observational data*. Philadelphia: Society for Industrial and Applied Mathematics.
- Wood, S. N. (2006). *Generalized additive models: An introduction with R*. Boca Raton: Chapman & Hall.